

(see Figure 17.28). In this approach, the existing defective gene is edited to correct the mutation.

➔ **Mastering Biology HHMI Video: Jennifer Doudna: CRISPR Provides a Tool for Changing Genes**



In 2018, researchers reported promising results using the CRISPR-Cas9 system in an attempt to correct the genetic defect that causes sickle-cell disease. They edited cells from patients with sickle-cell disease and injected them into bone marrow in mice. After 19 weeks, the gene remained corrected in 20–40% of the injected cells. Researchers, physicians, and patients alike are excited about the potential of CRISPR technology to treat or even cure human diseases that have a genetic basis, such as sickle-cell, Alzheimer's, and Parkinson's diseases, as well as some types of cancer. However, there are still concerns about using CRISPR-treated cells in humans because of possible effects on genes that are not being targeted; this is an active area of research.

In addition to technical challenges, gene therapy and gene editing provoke ethical questions. Some critics believe that tampering with human genes in any way is unethical. Other observers see no fundamental difference between transplanting genes into somatic cells and transplanting organs from one person to another.

An even more pressing question is the issue of engineering human germ-line cells to try to correct a defect in future generations. Such genetic engineering is now routinely done in laboratory mice, and some methods for genetic engineering of human embryos have been developed.

The development of the CRISPR-Cas9 system has engendered much debate about the ethics of gene editing. Jennifer Doudna (Figure 20.14), a co-discoverer of CRISPR-Cas9, and other biologists have agreed to exercise extreme caution as the field moves forward. Together, they called for the research community to “strongly discourage” any experimental work on human eggs or embryos.

In spite of this general consensus among biologists, a Chinese researcher reported in 2018 that he had used the CRISPR-Cas9 system to edit genes in embryos that completed fetal development and were born as twins and a third individual. He claimed to have edited the *CCR5* gene, which codes for a co-receptor for HIV (see Figure 7.8), so that HIV would be unable to bind and infect the cells. The twins' father was HIV-positive, which was the rationale the scientist used for carrying out this genetic engineering. However, this act was roundly condemned as highly unethical by the biological community; the researcher lost his job and may face a criminal investigation. In 2019, an advisory panel to the World Health Organization proposed establishing a registry to monitor any gene-editing research in humans.

There are many potential technical problems with the CRISPR-Cas9 technique, like the off-target effects mentioned

previously. Beyond those technical concerns, however, are underlying ethical considerations: Under what circumstances, if any, should genomes of human germ lines be altered? Would alterations lead to the practice of eugenics, a deliberate effort to control the genetic makeup of human populations? It is imperative to consider these questions.

Pharmaceutical Products

The pharmaceutical industry derives significant benefit from advances in DNA technology and genetic research, applying them to the development of useful drugs to treat diseases. Pharmaceutical products are synthesized using methods of either organic chemistry or biotechnology, depending on the nature of the product.

Synthesis of Small Molecules for Use as Drugs

Determining the sequence and structure of proteins crucial for tumor cell survival has led to the identification of small molecules that combat certain cancers by blocking the function of these proteins. One drug, imatinib (trade name Gleevec), is a small molecule that inhibits one tyrosine kinase (see Figure 11.8). The overexpression of this kinase, resulting from a chromosomal translocation, is instrumental in causing chronic myelogenous leukemia (CML; see Figure 15.16). Patients in the early stages of CML who are treated with imatinib have exhibited nearly complete, sustained remission from the cancer. Drugs that work in a similar way have also been used with success to treat a few types of lung and breast cancers. This approach is feasible only for cancers whose molecular basis is fairly well understood.

➔ **Mastering Biology HHMI Video: Gleevec**



In many cases of such drug-treated tumors, though, cells later arise that are resistant to the new drug. In one study, the whole genome of the tumor cells was sequenced both before and after the appearance of drug resistance. Comparison of the sequences showed genetic changes that allowed the tumor cells to “get around” the drug-inhibited protein. Here, we can see that cancer cells demonstrate the principles of evolution: Certain tumor cells have a random mutation that allows them to survive in the presence of a particular drug, and as a consequence of natural selection in the presence of the drug, these are the cells that survive and reproduce.

Protein Production in Cell Cultures Pharmaceutical products that are proteins are commonly synthesized on a large scale using cell cultures. You learned earlier in the chapter about DNA cloning and gene expression systems for producing large quantities of a chosen protein that is present naturally in only minute amounts. The host cells used in such expression systems can even be engineered to secrete a protein as it is made, thereby simplifying the task of purifying it by traditional biochemical methods.